Basics of Logic

Logic is a means of analysing logical statements.

Logical statement ↔ premise ↔ proposition

For example: proposition: Walter is a happy person

correct
true
applicable
logic 1

incorrect
not true
not applicable
logic 0

A **proposition** can be either correct (i.e. **true T**) or incorrect (i.e **false F**)

Several propositions may be connected via logical connectives. These include:

- Negation NOT
- Conjunction AND
- Inclusive disjunction OR
- Exclusive disjunction XOR

Truth Tables

Logical variables may assume one of two values true (T) or false (F).

One logical variable has 2 different values: $2^1 = 2$

Two logical variables have 2*2 different values: $2^2 = 4$

[redacted]

....

n different variables have $\underbrace{2 * 2 * \cdots * 2}_{\text{n times}}$ different values: 2^n

Examples: Truth Tables

Truth tables for 1 to 3 logical variables

p
T
F

p	q
T	T
T	F
F	T
F	F

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Logical Connectives

Operation	Meaning	Logic	Computer Science	Priority
Negation	NOT	$\neg$	! or ~	1
Conjunction	AND	$\wedge$	&	2
Inclusive disjunction	OR	$\vee$	|	3
Exclusive disjunction	XOR	$\oplus$	X	3
Implication	IMPLIES	$\Rightarrow$	$\rightarrow$	4
Equivalence	EQUIVALENCE	$\Leftrightarrow$	$\leftrightarrow$	4

Equivalence

P1: Descartes is human

P2: Descartes is mortal

Both premises are true, but they are not equivalent.

Descartes may be my cat. Then

- The cat is mortal but not human

P1: Descartes is human

P2: Descartes is a man

Both premises are true. But

You do not have to be a man to be human but you have to be human to be a man.

Being a man is **sufficient** for being human, but it is not **necessary**. The above statement could refer to Madame Descartes, who is not a man, but she is still human!

Equivalence

Have two propositions p and q.

p implies q then

- p is a **sufficient** condition for q
- q is a **necessary** condition for p

If p is both, a <u>necessary and a sufficient condition</u> for q then p and q are **equivalent**.

Then the terminology **“if and only if”** or also **iff** will be used.

Example: Descartes is married if and only if he has a wife.
p: *Descartes is married* and *Descartes has a wife* are equivalent.

Equivalence Truth Table

p	q	p ↔ q
T	T	T
T	F	F
F	T	F
F	F	T

Equivalence in the normal sense means: p and q are the same. The fact that p is true but q is false indicates that p is not equivalent to q. Therefore the statement, p is equivalent to q, p ↔ q, is not correct, i.e. it is false.

Implication

p: Descartes is human q: Descartes is mortal

The statement that Descartes is human implies that he is also mortal, as all humans are also mortal.

Example for mathematics: if $n \in \mathbb{N}$ then $n^2 \in \mathbb{N}$ but when $n \notin \mathbb{N}$ then $n^2 \notin \mathbb{N}$

But note:

if the premise is not correct, e.g. $\mathrm{n} = 1 + i \in \mathbb{N}$ then $n^2 \notin \mathbb{N}$

if the premise is not correct, e.g. $\mathrm{n} = -1$ with $n \notin \mathbb{N}$ but $n^2 \in \mathbb{N}$

A false premise may result in either a correct or an incorrect

Implication Truth Table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Negation

Negation reverse the logical sense:
True becomes false
False becomes true

p	¬ p
T	F
F	T

Conjunction

Conjunction is the statement for the logical AND. It means *and at the same time*.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Conjunction

p	q	$p \wedge q$	r	$(p \wedge q) \wedge r$
T	T	T	T	T
T	T	T	F	F
T	F	F	T	F
T	F	F	F	F
F	T	F	T	F
F	T	F	F	F
F	F	F	T	F
F	F	F	F	F

Conjunction

p	q	r	$q \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	T	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

Conjunction

p	q	r	p ∧ q ∧ r
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	F
F	F	F	F

Inclusive Disjunction

The inclusive disjunction, $p \vee q$, is true when either p, or q, or both of them, are true. Otherwise it is false.

p	**q**	**$p \vee q$**
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive Disjunction

The exclusive disjunction, p ⊕ q, is true when either p, or q, but if both of them are true, then p ⊕ q is false. Otherwise it is false.

p	q	p ⊕ q
T	T	F
T	F	T
F	T	T
F	F	F

Idempotence

Idempotence means: "*having the same power*" or "*not affected by*".
Consider proposition p and $p \vee p$

p	**$p \vee p$**	**$p \leftrightarrow p \vee p$**
T	T	T
F	F	T

Remark 1: the expression $p \leftrightarrow p \vee p$ reads p is equivalent to p or p. The last column then checks whether the result (T or F) is the same for both p and $p \vee p$.
Remark 2: A **logical statement** for which the result is **always either T or F** irrespective of the result of the individual input is a **tautology**.

Commutativity

Commutativity is connected with the **change of order**.
For example: $x \cdot y = y \cdot x$ and $5 \cdot 8 = 8 \cdot 5$.
Study

p	q	p ∨ q	q ∨ p	p ∨ q ↔ q ∨ p
T	T	T	T	T
T	F	T	T	T
F	T	T	T	T
F	F	F	F	T

Associativity

Associativity is connected with the order in which things are combined.

For example:

$$x \cdot y \cdot z = x \cdot (y \cdot z) = (x \cdot y) \cdot z$$

and

$$1 + 2 + 3 = 1 + (2 + 3) = (1 + 2) + 3$$

Associativity

p	q	r	$q \wedge r$	$p \wedge (q \wedge r)$	$p \wedge q$	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r) \leftrightarrow (p \wedge q) \wedge r$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	T
T	F	T	F	F	F	F	T
T	F	F	F	F	F	F	T
F	T	T	T	F	F	F	T
F	T	F	F	F	F	F	T
F	F	T	F	F	F	F	T
F	F	F	F	F	F	F	T

Distributivity

Distributivity is associated with the elimination of brackets.

For example:

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

and

$$2 \cdot (3 + 4) = 2 \cdot 7 = 2 \cdot 3 + 2 \cdot 4 = 6 + 8 = 14$$

Study the concept of distributivity in the context of logical statements for example

$$p \wedge (q \vee r) \leftrightarrow (p \wedge q) \vee (p \wedge r)$$

Associativity

p	q	r	$q \lor r$	$p \land (q \lor r)$	$p \land q$	$p \land r$	$(p \land q) \lor (p \land r)$	$p \land (q \lor r) \leftrightarrow (p \land q) \lor (p \land r)$
T	T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T	T
T	F	T	T	T	F	T	T	T
T	F	F	F	F	F	F	F	T
F	T	T	T	F	F	F	F	T
F	T	F	T	F	F	F	F	T
F	F	T	T	F	F	F	F	T
F	F	F	F	F	F	F	F	T

Associativity

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$	$p \vee (q \wedge r) \leftrightarrow (p \vee q) \wedge (p \vee r)$
T	T	T	T	**T**	T	T	**T**	T
T	T	F	F	**T**	T	T	**T**	T
T	F	T	F	**T**	T	T	**T**	T
T	F	F	F	**T**	T	T	**T**	T
F	T	T	T	**T**	T	T	**T**	T
F	T	F	F	**F**	T	F	**F**	T
F	F	T	F	**F**	F	T	**F**	T
F	F	F	F	**F**	F	F	**F**	T